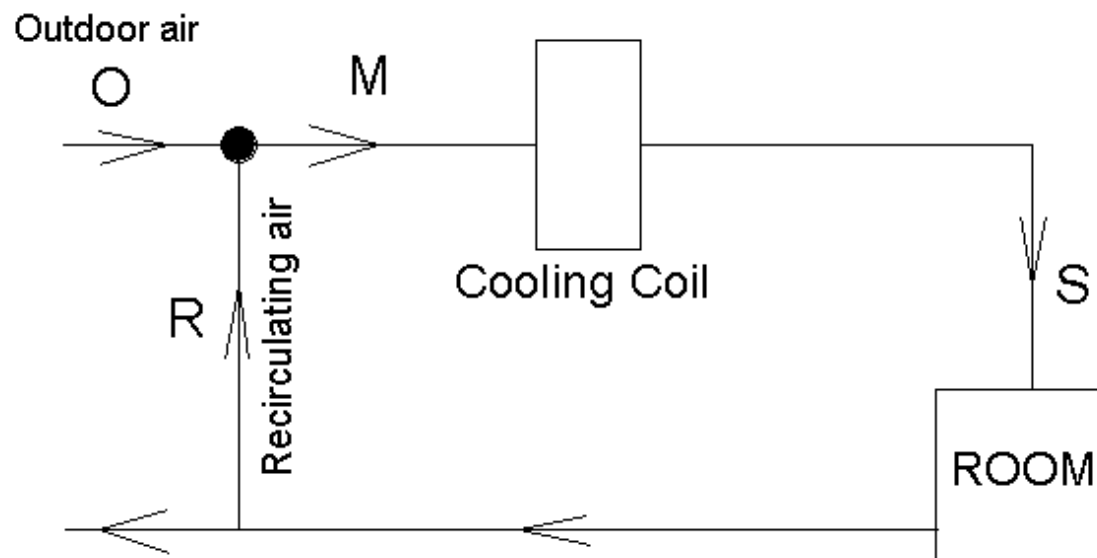
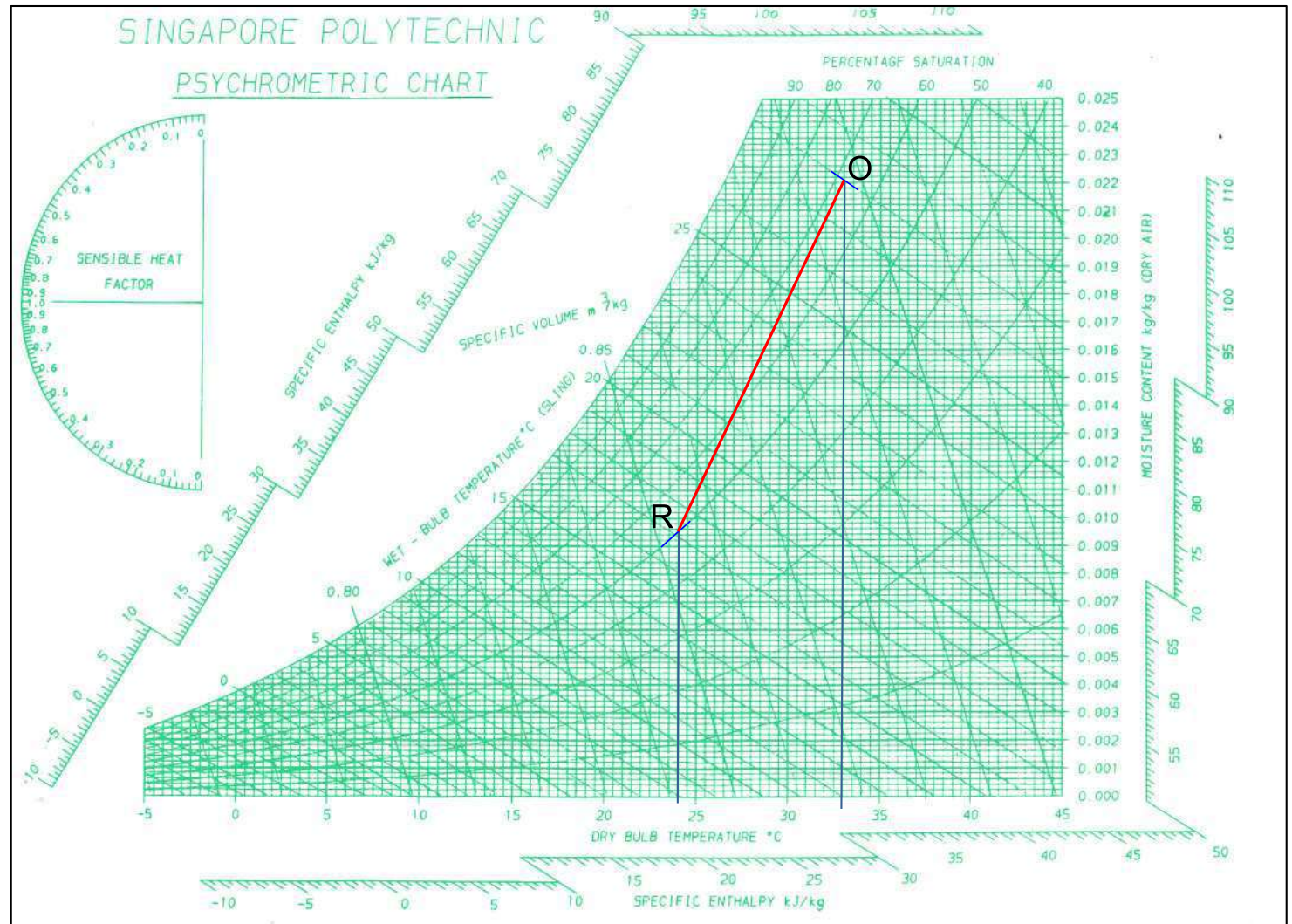
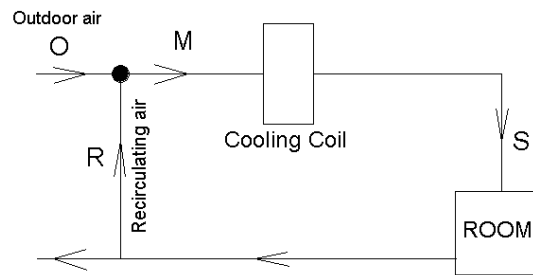


- (1) A room is maintained at  $24^{\circ}\text{C}$  db, 50% RH by an air handling system as shown below. The cooling coil has an apparatus dew point temperature of  $7^{\circ}\text{C}$ . The room has a sensible heat of 24 kW. The unit draws in  $0.5 \text{ m}^3/\text{s}$  of outdoor air. The outdoor air conditions are  $33^{\circ}\text{C}$  db,  $28^{\circ}\text{C}$  wb. If the mass flow rate of the air supplied to the room is  $1.99 \text{ kg/s}$ , determine:
- the mass flow rate of the outdoor air at point O (in kg/s) (0.557 kg/s),
  - the mass flow rate of the re-circulating air at point R (in kg/s) (1.433 kg/s),
  - the mixed air dry bulb temperature at point M ( $26.5^{\circ}\text{C}$ ),
  - the supply air dry bulb temperature at point S ( $12^{\circ}\text{C}$ ),
  - the bypass factor of the cooling coil (0.256),
  - the cooling coil load (in kW) (56.12 kW).





$$(a) \quad \dot{m}_O = \frac{\dot{V}_O}{v_O} = \frac{0.5}{0.898} = 0.557 \text{ kg/s}$$

$$(b) \quad \dot{m}_S = 1.99 \text{ kg/s}$$

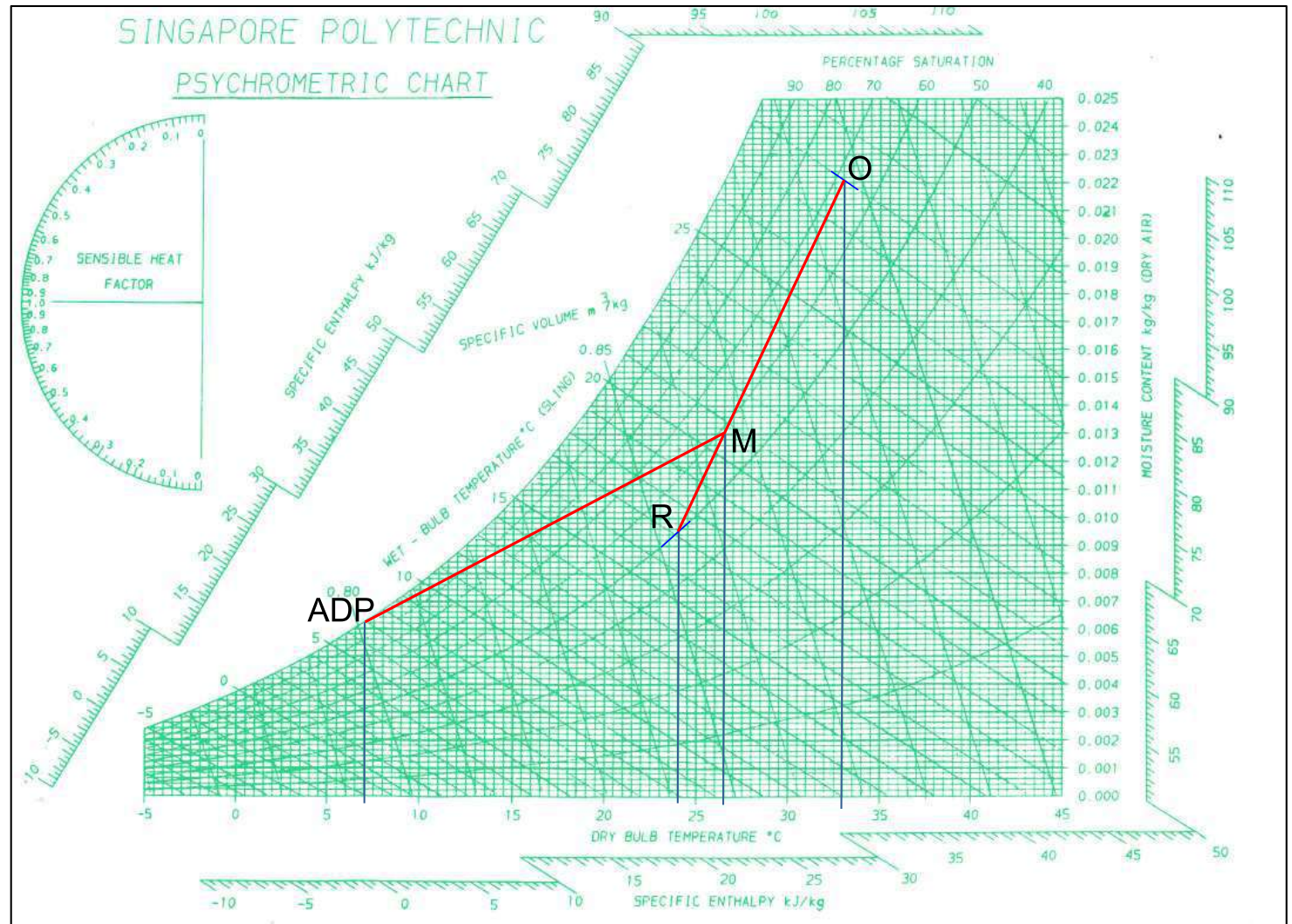
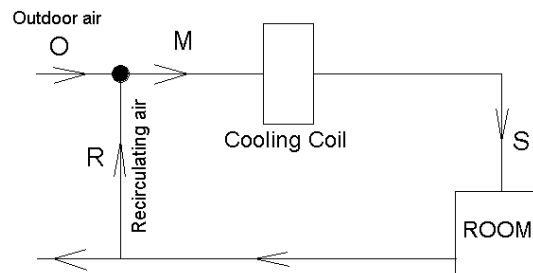
$$\dot{m}_R = 1.99 - 0.557 = 1.433 \text{ kg/s}$$

(c) *By mixing air equation for point M*

$$\frac{\dot{m}_O}{\dot{m}_S} = \frac{t_R - t_M}{t_R - t_O}$$

$$\frac{0.557}{1.99} = \frac{24 - t_M}{24 - 33}$$

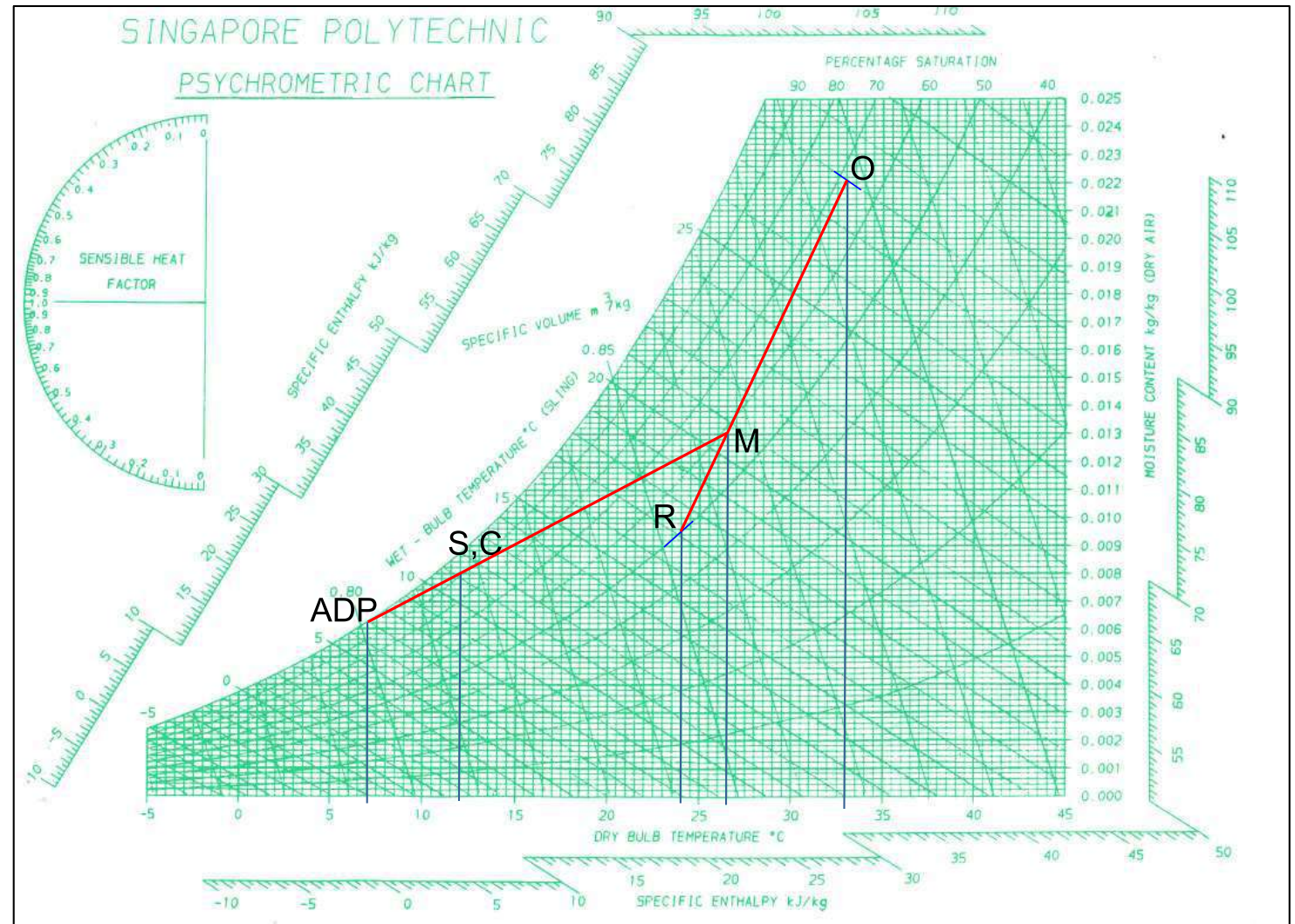
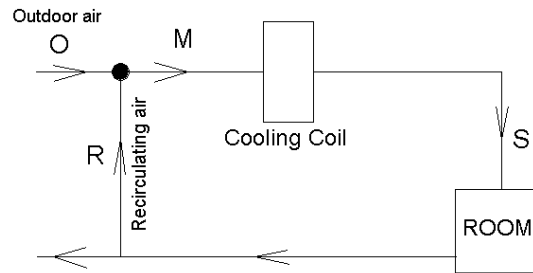
$$t_M = 26.5 \text{ }^\circ\text{C}$$



$$(d) \quad Q_{SH} = \dot{m}_S C_P (t_R - t_S)$$

$$24 = 1.99 \times 1.01 \times (24 - t_S)$$

$$t_S = 12 \text{ }^\circ\text{C}$$



(d)  $Q_{SH} = \dot{m}_S C_P (t_R - t_S)$

$$24 = 1.99 \times 1.01 \times (24 - t_S)$$

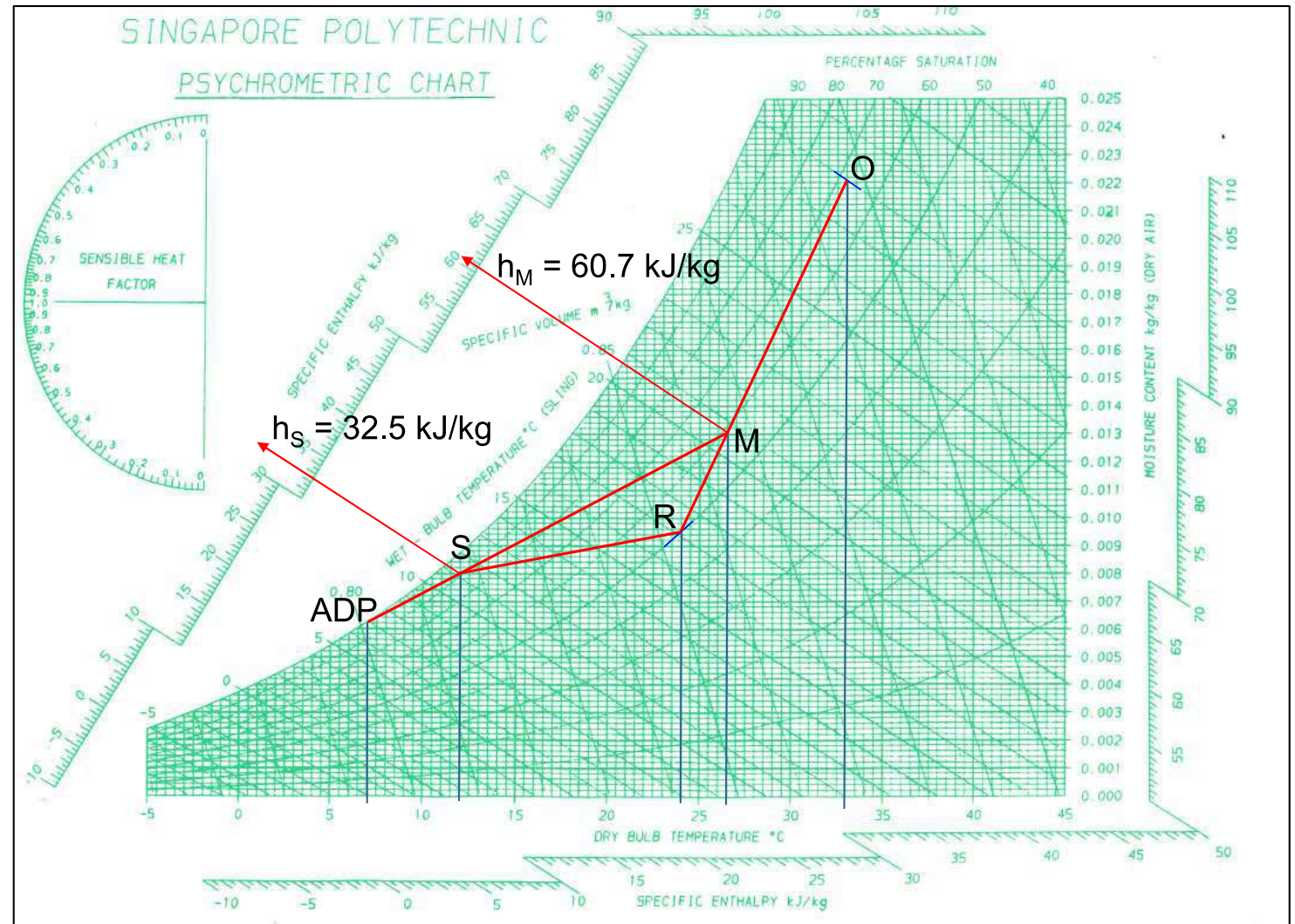
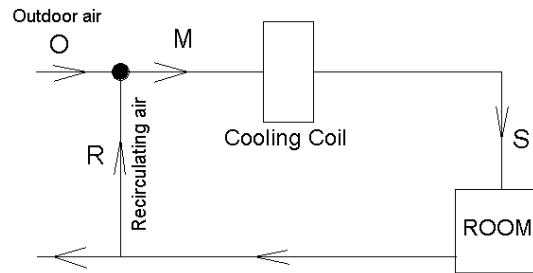
$$t_S = 12 \text{ }^\circ\text{C}$$

(e) *By using bypass factor equation*

$$BF = \frac{t_C - t_{ADP}}{t_M - t_{ADP}}$$

$$BF = \frac{12 - 7}{26.5 - 7}$$

$$BF = 0.256$$



$$(d) \quad Q_{SH} = \dot{m}_S C_P (t_R - t_S)$$

$$24 = 1.99 \times 1.01 \times (24 - t_S)$$

$$t_S = 12 \text{ }^\circ\text{C}$$

$$(e) \quad \text{By using bypass factor equation}$$

$$BF = \frac{t_C - t_{ADP}}{t_M - t_{ADP}}$$

$$BF = \frac{12 - 7}{26.5 - 7}$$

$$BF = 0.256$$

$$(f) \quad Q_{CC} = \dot{m}_S (h_M - h_S)$$

$$Q_{CC} = 1.99 \times (60.7 - 32.5)$$

$$Q_{CC} = 56.118 \text{ kW}$$